

Mahmoud Tarek Elkady

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Work Experience:

- **XR Developer at Merse (January 2024 - Present)**
 - Developed immersive XR experiences using Unity3D and specialized XR tools (Auto Hand - XR Interaction Toolkit).
 - Collaborated with multidisciplinary teams to design and implement XR solutions for training simulations.
 - Utilized cutting-edge XR technologies to enhance interactive storytelling and user experience design.
 - Developed reusable time system to replace Unity's time system such as coroutines for better control and optimization.
 - Conducted user testing and gathered feedback to iterate on XR experiences and optimize performance.
- **Part-time Senior XR Engineer at Bluebell (January 2026 - Present)**
 - Designed and implemented advanced XR systems utilizing Unity3D, ensuring high-performance interactive experiences.
 - Engineered reusable interaction mechanics and leveraged cutting-edge XR tools to streamline the development pipeline.
 - Collaborated with cross-functional teams to architect and deliver scalable solutions for complex simulations.
- **Game Developer at Ekson Almotahida Education Group (Apr 2023 - January 2024)**
 - Designed and developed educational simulations and mini-games from concept to implementation for scientific virtual labs.
 - Developed interactive simulations from scratch, and enhanced learning experiences through innovative game-based solutions using Unity and Unreal engine.
- **Game Development Instructor at SIMPLIFE & GOMYCODE (April 2023 - January 2024)**
 - Created comprehensive 2D/3D development curriculums covering game design, Unity scripting, and AR/VR integration.
 - Mentored students through collaborative hands-on projects from initial ideation to final deployment.

Education:

- **Postgraduate Diploma in Game Development (Oct 2021 - Jan 2022) - Information Technology Institute (ITI)**
- **Bachelor's Degree in Computer Science (2017 - 2021) - Misr International University (MIU)**
 - Implemented Deep Reinforcement Learning techniques in AR games to enhance user engagement and NPC behavior in augmented reality games. [Published](#) in IEEE - (Dec 2020 - May 2021)

Technical Skills:

- **Programming Languages:** C# - C++ - Python - Java - JavaScript.
- **Technologies:** Unity - Unreal Engine - Game Maker Studio - OpenGL - WebGL - Phaser.
- **Concepts:** OOP - ECS - MVC - Data Structures - Algorithms - Linear Algebra - Debugging.
- **Version Control & Collaboration:** Git - Jira - Confluence - Notion.

Notable Projects:

- **Augmented Reality Multiplayer RPG (MAHO) (Unity C#) - (Graduation project)**
 - Developed an AR Multiplayer RPG with Unity and Google's AR APIs. (AR-Foundation, AR Core & AR Kit)
 - Utilized Deep Reinforcement Learning for humanlike NPC behavior.
 - Implemented network functionality using Mirror Networking API.
- **2D MMORPG (Unity C#)**
 - Developed a 2D MMORPG with player mechanics and designed NPC monster AI with a comprehensive spawning system.
 - Ensured full network synchronization using Unity's Mirror Networking API.
- **Farm Invaders (Unity C#)**
 - Developed "Farm Invaders" using Unity and C# within a 2-day game jam, creating a 2D Survival Game.
 - Implemented a farming system, shooting mechanics with multiple weapons, a purchase system, and a wave spawning system.
- **Fingerprint Recognition (Python)**
 - An Image Processing project developed using Python that takes an input image from the user and compares it with a bench-marked dataset using image recognition techniques.



Extracurricular Activities:

- **E-Sports Coach at Sky Coach - (Oct 2019 - Mar 2020)**
 - Coached and interacted with students in the E-sport game *League of Legends*.
- **Pro Player in League of Legends**
 - Team Wild Panthers in the Greek League (Apr 2018 - Oct 2018).
 - Team Anubis Egyptian League (Aug 2019 - Sep 2019).
- **Member of Gamers Legacy Club at MIU (2018 - 2021)**
 - Organized gaming and coding events in MIU and Egycon.